Stat 240 Midterm

February 2026

AUTHOR
Peter Khrissanov 301611434

PUBLISHED
February 25, 2026

```r
download.file("https://owenward.github.io/files/nhl_data.tsv", "nhl_data.tsv")
read.csv("nhl_data.tsv", sep = "\t")
```

```
   Team GP SOG  W  L OTL PTS Reg_PTS       PPG SO_PTS    Pct_SO RW ROW SOW SOL
1    LA 82   9 47 25  10 104      98 1.3424658      6 0.6666667 37  41   6   3
2   STL 82   6 37 38   7  81      78 1.0263158      3 0.5000000 27  34   3   3
3   EDM 82   4 50 23   9 109     109 1.3974359      0 0.0000000 45  50   0   4
4   CGY 82   7 38 27  17  93      91 1.2133333      2 0.2857143 31  36   2   5
5   PIT 82   2 40 31  11  91      90 1.1250000      1 0.5000000 31  39   1   1
6   DET 82   7 35 37  10  80      77 1.0266667      3 0.4285714 28  32   3   4
7   PHI 82   3 31 38  13  75      73 0.9240506      2 0.6666667 26  29   2   1
8   WPG 82   2 46 33   3  95      94 1.1750000      1 0.5000000 36  45   1   1
9   SEA 82   4 46 28   8 100     100 1.2820513      0 0.0000000 37  46   0   4
10  CBJ 82   2 25 48   9  59      58 0.7250000      1 0.5000000 15  24   1   1
11  VAN 82   8 38 37   7  83      77 1.0405405      6 0.7500000 24  32   6   2
12  CHI 82   4 26 49   7  59      57 0.7307692      2 0.5000000 18  24   2   2
13  TOR 82   3 50 21  11 111     110 1.3924051      1 0.3333333 42  49   1   2
14  OTT 82   3 39 35   8  86      84 1.0632911      2 0.6666667 31  37   2   1
15  CAR 82   7 52 21   9 113     109 1.4533333      4 0.5714286 39  48   4   3
16  COL 82   9 51 24   7 109     103 1.4109589      6 0.6666667 36  45   6   3
17  FLA 82   3 42 32   8  92      90 1.1392405      2 0.6666667 36  40   2   1
18   SJ 82   7 22 44  16  60      59 0.7866667      1 0.1428571 16  21   1   6
19  MIN 82  13 46 25  11 103      96 1.3913043      7 0.5384615 34  39   7   6
20  DAL 82   7 47 21  14 108     104 1.3866667      4 0.5714286 39  43   4   3
     GF  GA DIFF    CONF
1   280 257   23 Western
2   263 301  -38 Eastern
3   325 260   65 Western
4   260 252    8 Western
5   262 264   -2 Eastern
6   240 279  -39 Western
7   222 277  -55 Eastern
8   247 225   22 Western
9   289 256   33 Western
10  214 330 -116 Eastern
11  276 298  -22 Western
12  204 301  -97 Western
13  279 222   57 Eastern
14  261 271  -10 Eastern
15  266 213   53 Eastern
16  280 226   54 Western
17  290 273   17 Eastern
```

```
18 234 321  -87 Eastern
19 246 225   21 Western
20 285 218   67 Western
```

```
library(RSQLite)
library(DBI)
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.6
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.1     ✔ tibble    3.3.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.2
✔ purrr     1.2.1
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to
become errors
```

Problem 1

#a

```
## write code here
data <- read.csv("nhl_data.tsv", sep = "\t")
nhl_sub <- data |> filter(W > 50)
nhl_sub
```

```
  Team GP SOG  W  L OTL PTS Reg_PTS      PPG SO_PTS    Pct_SO RW ROW SOW SOL
1  CAR 82   7 52 21   9 113     109 1.453333      4 0.5714286 39  48   4   3
2  COL 82   9 51 24   7 109     103 1.410959      6 0.6666667 36  45   6   3
   GF  GA DIFF    CONF
1 266 213   53 Eastern
2 280 226   54 Western
```

```
print(dim(nhl_sub)[1])
```

```
[1] 2
```

Problem 1

#b

```
nhl_mod <- select(nhl_sub, -GF, -GA, -DIFF)
nhl_mod
```

```
   Team GP SOG  W  L OTL PTS Reg_PTS      PPG SO_PTS    Pct_SO RW ROW SOW SOL
1   CAR 82   7 52 21   9 113     109 1.453333      4 0.5714286 39  48   4   3
2   COL 82   9 51 24   7 109     103 1.410959      6 0.6666667 36  45   6   3
     CONF
1 Eastern
2 Western
```

Problem 1

#c

```
write.table(nhl_mod, file = "nhl.psv", sep="._.")
```

Problem 2

#a

```
## write code here

db <- dbConnect(SQLite(), "nhl_data.sqlite")
data <- read.csv("nhl_data.tsv", sep = "\t")

dbWriteTable(db, "NHL", data, overwrite = TRUE)
```

Problem 2

#b

```
## write code here
object <- dbGetQuery(db, "SELECT * FROM NHL WHERE W > 50")
object
```

```
   Team GP SOG  W  L OTL PTS Reg_PTS      PPG SO_PTS    Pct_SO RW ROW SOW SOL
1   CAR 82   7 52 21   9 113     109 1.453333      4 0.5714286 39  48   4   3
2   COL 82   9 51 24   7 109     103 1.410959      6 0.6666667 36  45   6   3
   GF  GA DIFF    CONF
1 266 213   53 Eastern
2 280 226   54 Western
```

```
size <- print(dim(object)[1])
```

```
[1] 2
```

```
size
```

```
[1] 2
```

##Problem 2 #c

```
dbGetQuery(db,
           "SELECT CONF,
          COUNT(*) AS n,
          AVG(w) AS avg_wins
   FROM NHL
   GROUP BY CONF"
)
```

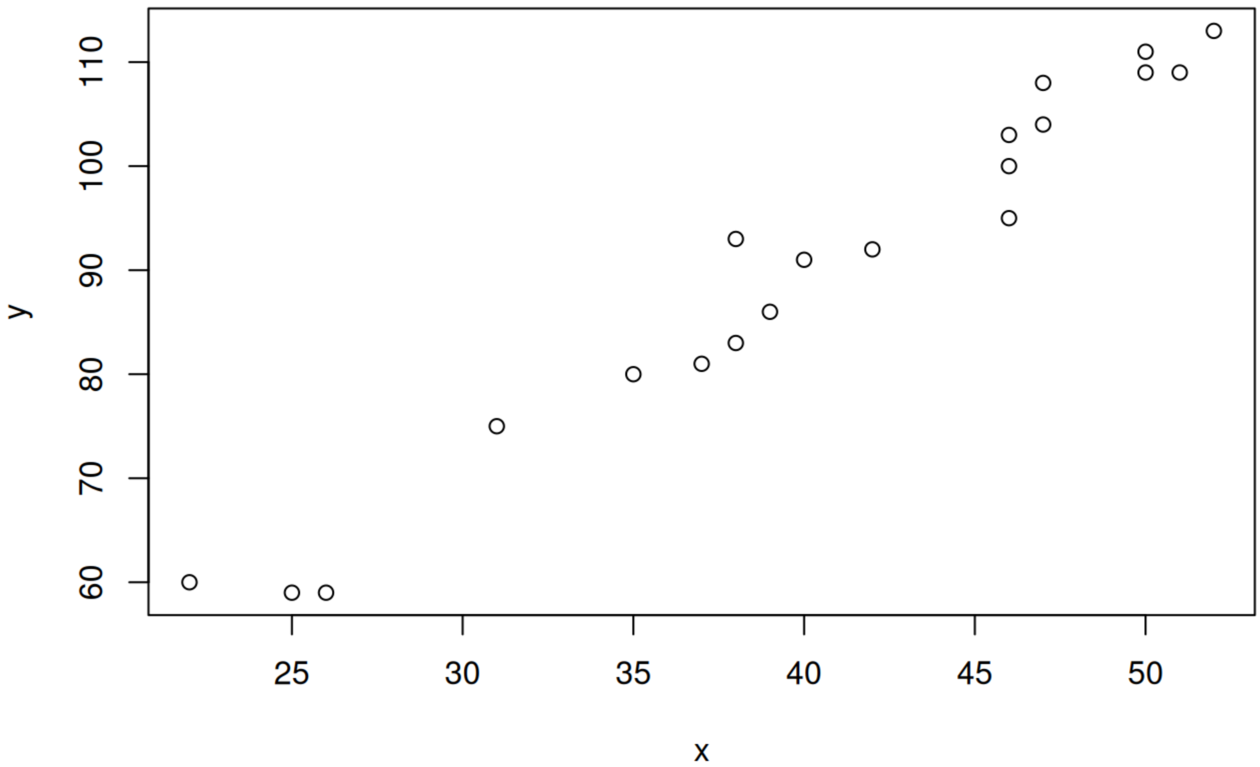
```
     CONF  n avg_wins
1 Eastern  9 37.55556
2 Western 11 42.72727
```

Problem 3 (Delete if not used)

a

```
## write code here
x <- data$W
y <- data$PTS

plot(x, y)
```



b

```
## write code here

model <- lm(y ~ x)
alpha <- summary(model)$coefficients[1, 1]
beta <- summary(model)$coefficients[2, 1]
print("alpha:")
```

```
[1] "alpha:"
```

```
alpha
```

```
[1] 13.47326
```

```
print("beta:")
```

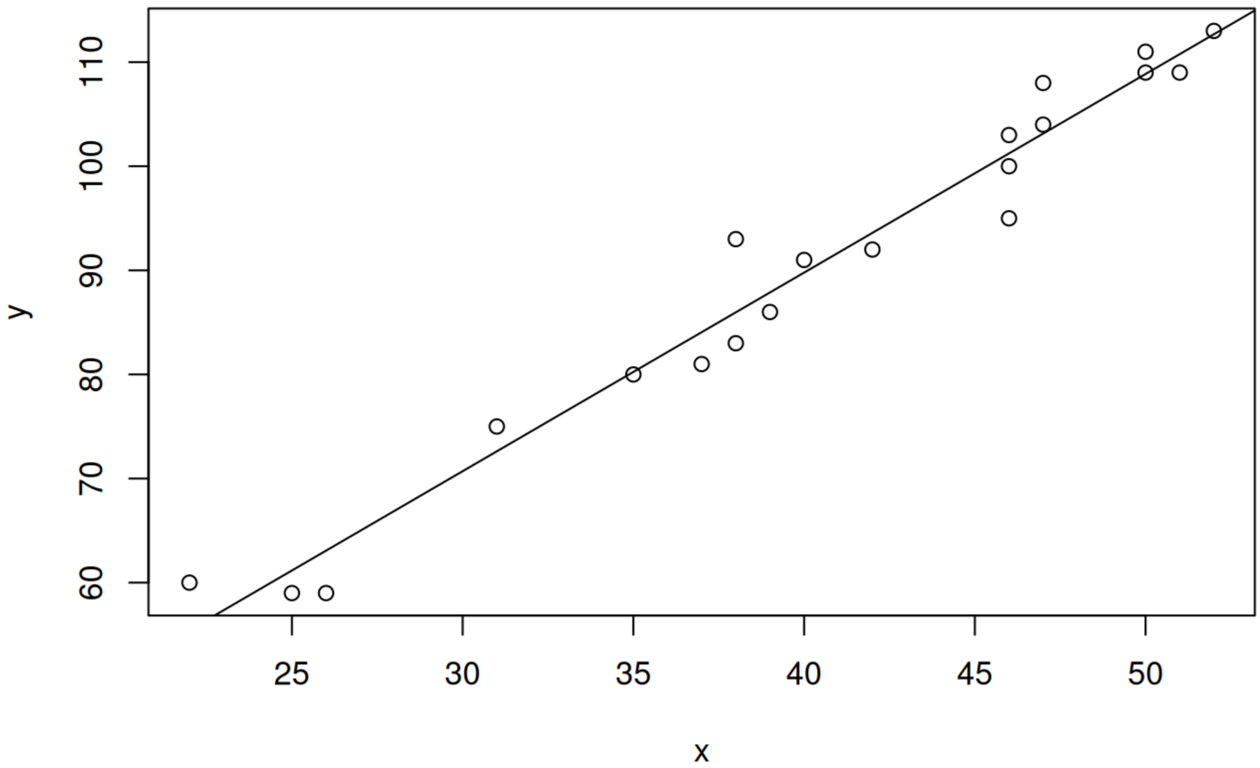
```
[1] "beta:"
```

```
beta
```

```
[1] 1.90784
```

```r
x <- data$W
y <- data$PTS

plot(x, y)
abline(a= alpha, b = beta)
```



C

```r
## write code here

SSE <- function(alpha, beta, x, y) {
  alpha <- alpha
  beta <- beta
  f_x <- alpha + beta * x
  SSE <- sum((y - f_x)^2)
  SSE
}

print(SSE(alpha, beta, x, y))
```

```
[1] 198.6633
```

```r
print(SSE(0, 0, x, y))
```

```
[1] 169793
```